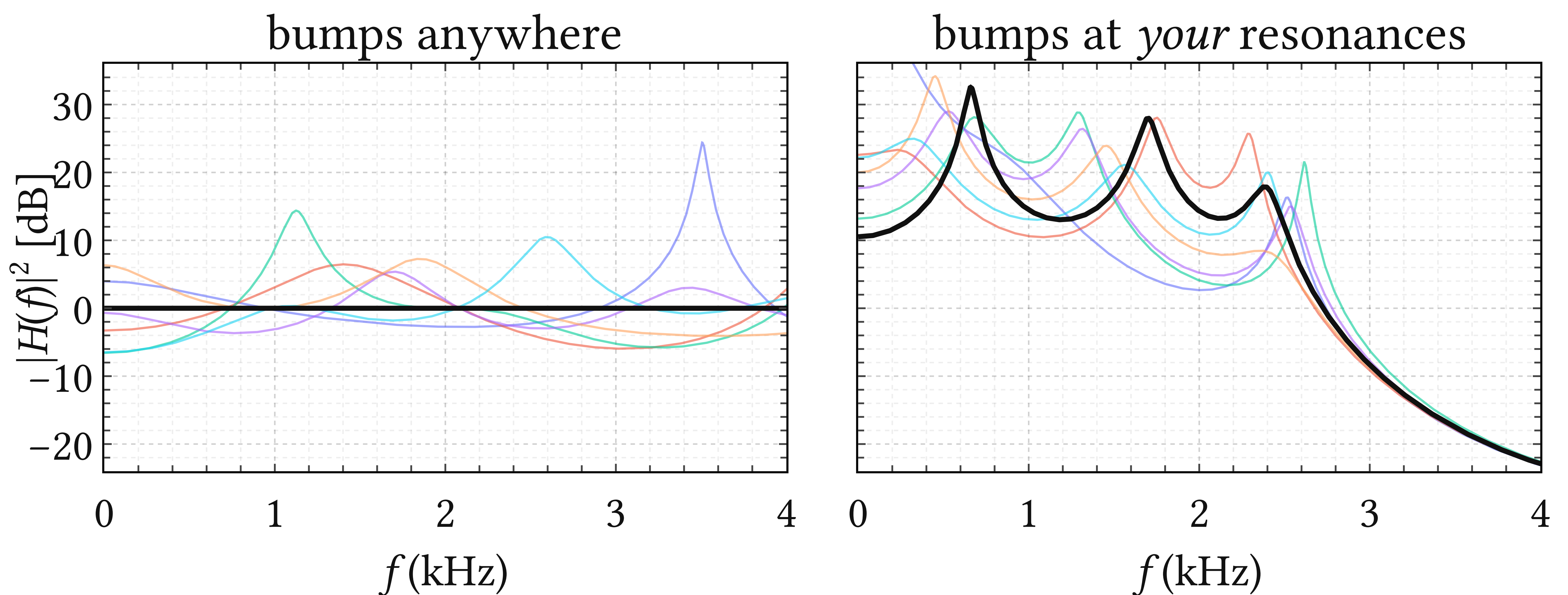


# WANTED

## ILL-POSED AUTOREGRESSIVE PROBLEMS



Envelope draws from a Gaussian AR(6) prior, before and after one closed-form  $D_{\text{KL}}$  update toward three formants (dotted); bold: prior mean.

### THE MODEL

AR( $P$ ) with a Gaussian prior on the coefficients is conjugate: the posterior stays closed form.

$$x_t = a_1 x_{t-1} + \dots + a_P x_{t-P} + \varepsilon_t$$

$$\mathbf{a} \sim \text{Normal}(\boldsymbol{\mu}, \mathbf{S})$$

### THE DICTIONARY

$A(z) = 1 - a_1 z^{-1} - \dots - a_P z^{-P}$  shapes the envelope  $|H|^2 = 1/|A|^2$ ; every expected feature is a factor:

| Spectral feature                | $Q(z)$                                               | deg  |
|---------------------------------|------------------------------------------------------|------|
| Resonance at $(\rho, \omega_0)$ | $1 - 2\rho \cos(\omega_0)z^{-1} + \rho^2 z^{-2}$     | 2    |
| Real pole at $\rho$             | $1 - \rho z^{-1}$                                    | 1    |
| Tilt $-6k$ dB/oct               | $(1 - \rho z^{-1})^k$                                | $k$  |
| Seasonal pole of period $s$     | $1 - z^{-s}$                                         | $s$  |
| $k$ -fold resonance             | $(1 - 2\rho \cos(\omega_0)z^{-1} + \rho^2 z^{-2})^k$ | $2k$ |

### WORKED EXAMPLE

AR(4), impose one resonance  $Q(z) \mid A(z)$ :

$$A(z) = 1 - a_1 z^{-1} - a_2 z^{-2} - a_3 z^{-3} - a_4 z^{-4}$$

$$Q(z) = 1 - cz^{-1} + dz^{-2}, \quad c = 2\rho \cos \omega_0, d = \rho^2$$

Divisibility is linear in  $\mathbf{a}$ :

$$Q \mid A \iff \mathbf{F}\mathbf{a} + \mathbf{c} = \mathbf{0}$$

$$\mathbf{F} = \begin{pmatrix} 1 & 0 & -1/d & -c/d^2 \\ 0 & 1 & c/d & c^2/d^2 - 1/d \end{pmatrix}$$

Impose it *on average*,  $\mathbb{E}_{\mathbf{a}}[\mathbf{F}\mathbf{a} + \mathbf{c}] = \mathbf{0}$ , by the minimum- $D_{\text{KL}}$  update of  $\text{N}(\mathbf{0}, \lambda \mathbf{I})$ : a Gaussian prior with the belief in its mean, uncertainty intact.

### REWARD

a prior with your knowledge in it, and a co-author who checks the math.

